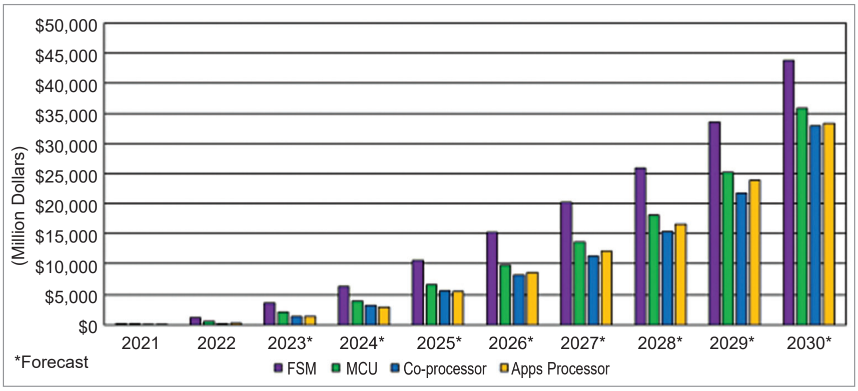

Fig. 3: RISC-V SoCs revenues by functionality category, 2021-2030 (Source: The SHD Group, January 2024)

licensing revenues by 2030. As the IPs start coming from different vendors and in bulk, licence revenues or fees (permitting use of an IP) start reaching a saturation point. Meanwhile, as the IP is used in bulk, profit rises, and so does royalty revenue (a fixed percentage of profit earned from an IP).

The shift in revenue mix is due to increased shipments of SoCs with CPU cores in later years. This shift indicates that more companies are using CPU cores in their products and paying royalties based on their usage, rather than just licensing the IP.

CPU IP royalties reached $16.7M in 2022 and are forecast to grow to $968.1M by 2030, a CAGR of 60.2%. IP royalties are payments made by a licensee (someone who is granted permission to use the intellectual property) to the licensor (the owner of the intellectual property) for the use of that IP. These payments are typically based on a percentage of the revenue generated by the licensee from products or services that incorporate the IP. Royalties are often paid on a regular basis, such as monthly or quarterly.

CPU IP licensing revenues were $59.1M in 2022 and are forecast to reach $542.9M by 2030, a CAGR of 26.8%. Licensing revenues refer to the total income generated by licensing intellectual property rights to third parties. This includes not only royalties but also any upfront fees, milestone payments, or other payments made as part of the licensing agreement. Licensing revenues are the total financial benefit received by the licensor from the licensing of their IP.

The RISC-V market is dynamic and evolving, with new companies entering and introducing powerful CPU IP types. Constant innovation and the introduction of new CPU cores push the ratio between royalty and licensing revenues higher for RISC-V.

Effect of AI. AI is significantly impacting silicon design and creating opportunities for RISC-V-based SoCs. Even with the current boost in chip manufacturing, the demand for AI chips is not being fulfilled. OpenAI CEO Sam Altman, who is looking to venture into chip manufacturing, recently said that current chip manufacturing is not able to withstand the growing AI adoption. The popularity of AI applications like ChatGPT is changing how computers are used and driving the demand for high-performance silicon. This demand presents a significant opportunity for RISC-V designs, especially in PCs, laptops, tablets, industrial, consumer, and automotive markets.

Recent advancements

China's reliance. China is leveraging the RISC-V architecture to counter US sanctions and advance its semiconductor industry, with significant investment in RISC-V projects and a mature ecosystem. RISC-V chips, increasingly used in various applications, are closing the performance gap with ARM chips. China's military and research institutions are developing RISC-V technologies, contributing to the country's push for technological self-sufficiency.

Quintauris. After RISE, Quintauris GmbH, a collaboration between major semiconductor companies like

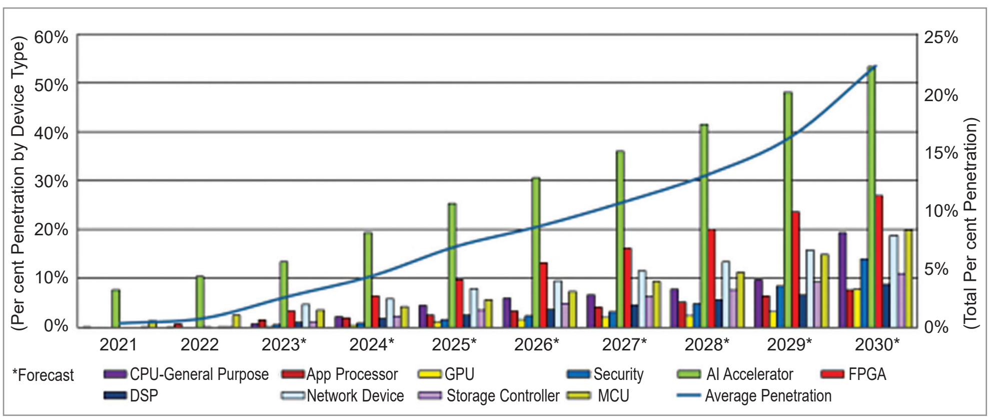

Fig. 4: Penetration of RISC-V SoCs by revenue of device types, 2021-2030 (Source: The SHD Group, January 2024)